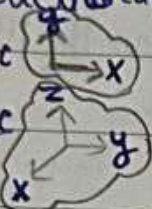


① $\mathbb{R}$ → the set of all real numbers

$\mathbb{R}^2, \mathbb{R}^3, \dots, \mathbb{R}^n$ → Represents
n-dimensional Euclidean spaces

$\mathbb{R}^2$ → 2D plane

$\mathbb{R}^3$ → 3D space



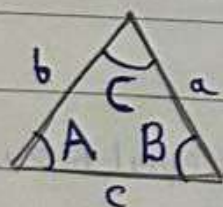
Linear Algebra

② $\|\vec{V}\|$
or $|\vec{V}|$

L2 Norm = $\sqrt{V_1^2 + V_2^2 + \dots + V_n^2}$
represents the vector's
magnitude (length)

③ (x, y) for unit circle is $(\cos\theta, \sin\theta)$

④



law of sines:

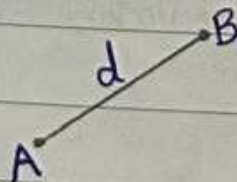
$$\frac{a}{\sin(A)} = \frac{b}{\sin(B)} = \frac{c}{\sin(C)}$$

law of cosines:

$$c^2 = a^2 + b^2 - 2ab \cos(C)$$

⑤ Distance Between Two points (Euclidean distance)

$$d = \sqrt{(a_1 - b_1)^2 + \dots + (a_n - b_n)^2}$$

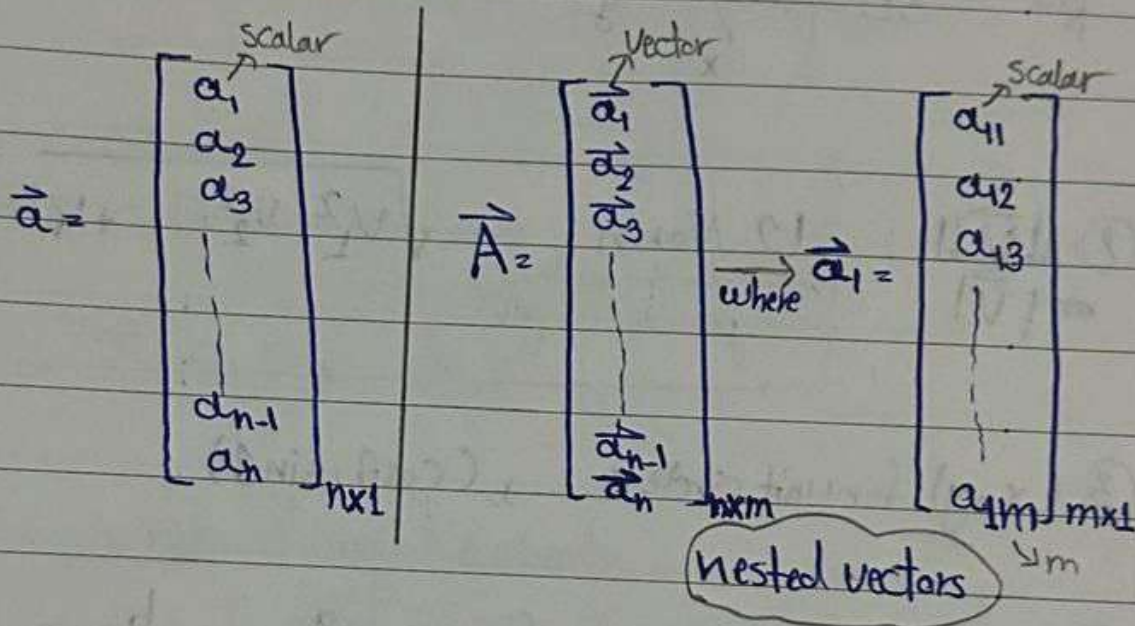


⑥ Scalar: a single numerical value, often representing ^①magnitude or quantity
vector: an ordered array of numbers, which can represent both ^①magnitude and ^②direction in space

⑦ indexing in vectors

- Standard mathematical notation indexes n -vectors from $i=1$ to $i=n$
- Ambiguity in notation: a_i could mean the i th element of a vector or the i th vector in a collection

ex 1- for $\mathbb{R}^n$



⑧ special vectors

1. Zero Vector $\rightarrow$ A vector with all elements equal to zero. For $\mathbb{R}^3$, $\begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$.
2. Unit Vector $\rightarrow$ Vector with a single element equal to one. For $\mathbb{R}^3$, $\begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}$ or $\begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix}$ or $\begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}$.
3. sparse vector $\rightarrow$ is characterized by having many of its entries as zero. Unit vector is sparse vector. must have the same size. For $\mathbb{R}^5$, $\begin{bmatrix} 3 \\ 0 \\ 0 \\ 4 \\ 0 \end{bmatrix}$.

⑨ $\vec{a} = \begin{bmatrix} 2 \\ 3 \end{bmatrix}$, $\vec{b} = \begin{bmatrix} 1 \\ 4 \end{bmatrix}$

Addition of vectors $\rightarrow \vec{a} + \vec{b} = \begin{bmatrix} 3 \\ 7 \end{bmatrix}$

subtraction of vectors $\rightarrow \vec{a} - \vec{b} = \begin{bmatrix} 1 \\ -1 \end{bmatrix}$

$\downarrow$
to visualize vector $\vec{a}$

$\rightarrow$ the result has the same size too

⑩ properties of vector Addition:

1. Commutative $\rightarrow a + b = b + a$

2. Associative $\rightarrow (a + b) + c = a + (b + c)$

3. Addition of zero vector $\rightarrow a + \vec{0} = \vec{0} + a = a$

4. Subtracting a vector from itself $\rightarrow a - a = \vec{0}$

$$\begin{bmatrix} 0 \\ 0 \\ \vdots \\ 0 \end{bmatrix}_{n \times 1} = \vec{0}$$

⑪ Scalar multiplication $\rightarrow$ multiplying each component of a vector by a scalar value, effectively scaling the vector's magnitude

ex: $\vec{a} = \begin{bmatrix} a_1 \\ a_2 \\ \vdots \\ a_n \end{bmatrix}$

$$\therefore c \cdot \vec{a} = \begin{bmatrix} c \cdot a_1 \\ c \cdot a_2 \\ \vdots \\ c \cdot a_n \end{bmatrix}_{n \times 1}$$

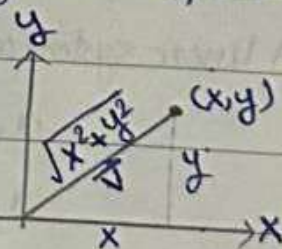
⑫ Dot product

• The dot product of a vector $\vec{V} = \begin{bmatrix} x \\ y \end{bmatrix}$ with itself gives the square of the length of $\vec{V}$

 $\|\vec{V}\|^2$

$$\vec{V} \cdot \vec{V} = x^2 + y^2 = \|\vec{V}\|^2$$

$$\vec{V} \cdot \vec{V} = x \cdot x + y \cdot y$$



• the dot product of two vectors a & b is:

$\downarrow$
gives us a measure of their similarity

$$\vec{a} \cdot \vec{b} = a_1 b_1 + a_2 b_2 + \dots + a_n b_n$$

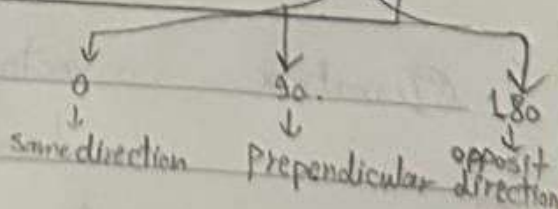
or $\vec{a}^T \vec{b}$

$$\vec{a} \cdot \vec{b} = \|\vec{a}\| \cdot \|\vec{b}\| \cdot \cos \theta$$

$$\cos \theta = \frac{\vec{a} \cdot \vec{b}}{\|\vec{a}\| \cdot \|\vec{b}\|}$$

$$-1 \leq \cos \theta \leq 1$$

$$\therefore \|\vec{a} \cdot \vec{b}\| \leq \|\vec{a}\| \cdot \|\vec{b}\|$$



note 1 - $\|V\| = 0$ only when $\rightarrow V$ is zero vector ($V = \vec{0}$)

التاريخ:

الموضوع:

⑬ inner product $\rightarrow$ generalizes the dot product to more abstract vector spaces, which can include spaces with complex numbers

• For complex vectors $\rightarrow$ the inner product involves conjugating one of the vectors

• For real vector spaces $\rightarrow$ the inner product is equivalent to the dot product

⑭ Cauchy - Schwarz Inequality

$$|\langle \vec{x}, \vec{y} \rangle| \leq \|\vec{x}\| \cdot \|\vec{y}\|$$

Linear system

⑮ A linear system with m equations & n unknowns:

$$a_{11}x_1 + a_{12}x_2 + \dots + a_{1n}x_n = b_1$$

$$a_{21}x_1 + a_{22}x_2 + \dots + a_{2n}x_n = b_2$$

$$\vdots$$

$$a_{m1}x_1 + a_{m2}x_2 + \dots + a_{mn}x_n = b_m$$

a_{ij}
row $\rightarrow$ column

⑯ Linear system $\rightarrow$ homogeneous $\rightarrow$ all constant terms b_i are zero
Nonhomogeneous $\rightarrow$ otherwise

⑰ matrix $\rightarrow$ a rectangular array of real numbers arranged in rows and columns

ex - $m \times n$ matrix A

$$A = \begin{pmatrix} a_{11} & a_{12} & \dots & a_{1n} \\ a_{21} & a_{22} & \dots & a_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{m1} & a_{m2} & \dots & a_{mn} \end{pmatrix}$$

no. of rows $\rightarrow m$
no. of columns $\rightarrow n$
 $m \times n$

18 Matrix operations

1. sum of two matrices A & B of the same dimensions:-

obtained by adding
their corresponding
elements

$$(A+B)_{ij} = A_{ij} + B_{ij}$$

2. the difference of two matrices A & B of the same dimensions:-

obtained by subtracting
their corresponding
elements

$$(A-B)_{ij} = A_{ij} - B_{ij}$$

3. Scalar Multiplication of a matrix A by a scalar α results in a new matrix where each entry of A is multiplied by α .

$$\text{ex: } \alpha = c \rightarrow c \cdot A = \begin{bmatrix} c \cdot a_{11} & c \cdot a_{12} \\ c \cdot a_{21} & c \cdot a_{22} \end{bmatrix}_{2 \times 2}$$

4. The product of an $m \times n$ matrix A and $n \times p$ matrix B results in an $m \times p$ matrix C, where each entry C_{ij} is computed as the dot product of the i th row of A and the j th column of B.

$$\text{ex: } A = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}_{2 \times 2}, B = \begin{bmatrix} 2 & 0 \\ 1 & 2 \end{bmatrix}_{2 \times 2}$$

$$AB = \begin{bmatrix} 1 \times 2 + 2 \times 1 & 1 \times 0 + 2 \times 2 \\ 3 \times 2 + 4 \times 1 & 3 \times 0 + 4 \times 2 \end{bmatrix}_{2 \times 2} \rightarrow AB = \begin{bmatrix} 4 & 4 \\ 10 & 8 \end{bmatrix}_{2 \times 2}$$

① Matrix representation of a linear system

$$\begin{array}{c}
 \text{A} \qquad \qquad \qquad \text{X} \qquad \qquad \qquad \text{B} \\
 \begin{bmatrix} a_{11} & a_{12} & \dots & a_{1n} \\ a_{21} & a_{22} & \dots & a_{2n} \\ \vdots & \vdots & & \vdots \\ a_{m1} & a_{m2} & \dots & a_{mn} \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{bmatrix} = \begin{bmatrix} b_1 \\ b_2 \\ \vdots \\ b_m \end{bmatrix} \\
 \begin{matrix} m \times n & n \times 1 & m \times 1 \end{matrix}
 \end{array}$$

$$AX = B$$

a specific structure of matrix defined by leading ones, zeros rows at the bottom, and a staircase pattern used to solve linear eqs

② Gaussian Elimination → transform the system into a row-echelon form, from which the solution can be more easily identified
systematic method

augmented coefficient matrix →

$$(A|b) = \begin{bmatrix} a_{11} & a_{12} & \dots & a_{1n} & b_1 \\ a_{21} & a_{22} & \dots & a_{2n} & b_2 \\ \vdots & \vdots & & \vdots & \vdots \\ a_{m1} & a_{m2} & \dots & a_{mn} & b_m \end{bmatrix}$$

dimension = $m \times (n+1)$

② Row Echelon Form (REF)

a matrix A is in row echelon form if:

1. All nonzero rows are above any rows of all zeros
 2. Each leading entry of a row is to the right of the leading entry of the row above it
 3. All entries in a column below a leading entry are zeros
- * The leading entry in each row is known as the pivot or corner entry

② Reduced Row Echelon Form (RREF)

A matrix is in RREF if, in addition to REF criteria:-

4- Every leading entry in a row is 1 (also called a leading 1)

5- Each leading 1 is the only nonzero entry in its column

• RREF simplifies the system for easier solution derivation

Ex:- $X_1 + X_2 + X_3 = 6$

$2X_1 + 3X_2 + X_3 = 14$

$X_1 - X_2 + 2X_3 = 8$

Step 1:- Augmented Matrix $\rightarrow$

$$A = \left[\begin{array}{ccc|c} 1 & 1 & 1 & 6 \\ 2 & 3 & 1 & 14 \\ 1 & -1 & 2 & 8 \end{array} \right]_{3 \times 4}$$

Step 2:- Get (RREF) of A

1- $\left. \begin{array}{l} R_2 = R_2 - 2R_1 \\ R_3 = R_3 - R_1 \end{array} \right\} \rightarrow A = \left[\begin{array}{ccc|c} 1 & 1 & 1 & 6 \\ 0 & 1 & -1 & 2 \\ 0 & -2 & 1 & 2 \end{array} \right]$

2- $\left. \begin{array}{l} R_3 = R_3 + 2R_2 \\ R_1 = R_1 - R_2 \end{array} \right\} \rightarrow A = \left[\begin{array}{ccc|c} 1 & 0 & 2 & 4 \\ 0 & 1 & -1 & 2 \\ 0 & 0 & -1 & 6 \end{array} \right]$

3- $R_3 = -1 * R_3 \rightarrow A = \left[\begin{array}{ccc|c} 1 & 0 & 2 & 4 \\ 0 & 1 & -1 & 2 \\ 0 & 0 & 1 & -6 \end{array} \right]$

4- $\left. \begin{array}{l} R_2 = R_2 + R_3 \\ R_1 = R_1 - 2R_3 \end{array} \right\} \rightarrow A = \left[\begin{array}{ccc|c} 1 & 0 & 0 & 16 \\ 0 & 1 & 0 & -4 \\ 0 & 0 & 1 & -6 \end{array} \right] \rightarrow \text{RREF}$

step 3 - rewrite the system

$$\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} X_1 \\ X_2 \\ X_3 \end{bmatrix} = \begin{bmatrix} 16 \\ -4 \\ -6 \end{bmatrix}$$

$3 \times 3 \quad \quad 3 \times 1$

$$\therefore X_1 = 16$$

$$\therefore X_2 = -4$$

$$\therefore X_3 = -6$$

② there is 3 Cases

Case 1 -

$$\left[\begin{array}{ccc|c} & & & 3 \\ & & & 4 \\ & & & 2 \\ 0 & & & 5 \end{array} \right] \rightarrow 0=5 \rightarrow \therefore \text{there is no solution}$$

Case 2 -

when number of unknowns > number of equations

$\rightarrow \therefore$ there is no unique solution (infinite number of solution)

Case 3 - Single solution